

PEP-C

PEP-C is a cancer drug treatment. It is used to treat lymphoma.

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What is PEP-C?

PEP-C is a combination of cancer drugs. It is used to treat advanced [lymphoma](#). It is a combination of a steroid with 3 chemotherapy drugs.

The name PEP-C comes from the initials of the drugs used:

- P – prednisolone is a [steroids](#)
- E – [etoposide](#) is a chemotherapy drug
- P – [procarbazine](#) is a chemotherapy drug
- C – [cyclophosphamide](#) is a chemotherapy drug.

It is best to read this information with our general information about [cancer drug treatments](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

Having PEP-C

All 4 drugs come as capsules or tablets. This means you can take them at home.

During your course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor or blood cancer doctor (haematologist)
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your course of chemotherapy

Your cancer team will discuss your treatment plan with you. They may give you a copy of the treatment plan to take home.

Your doctor, nurse or pharmacist will tell you whether to take the tablets every day or only on certain days each week. They will talk to you about which days of the week you should have treatment. It is important to take your treatment exactly as explained.

Each day you have treatment, you will have:

- prednisolone tablets – take after breakfast
- cyclophosphamide tablets – take after lunch
- etoposide capsules – take after dinner
- procarbazine capsules – take at bedtime.

Your cancer team will also give you anti-sickness drugs and other medicines to take home. Take these exactly as they tell you to.

Taking your tablets and capsules

Your cancer team will give you the tablets and capsules to take home. They may give you tablets or capsules of different strengths. Always take them exactly as explained. This is important to make sure they work as well as possible for you.

Make sure you:

- swallow each tablet or capsule whole with a glass of water
- do not chew, open, break or crush them.

If you forget to take any of the capsules or tablets, contact the hospital on the 24-hour number you have been given for advice. Never take a double dose to make up for a missed dose.

Other things to remember about your capsules and tablets:

- Wash your hands after taking them.
- Other people should avoid direct contact with them.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused capsules or tablets. Return them to your cancer team at the hospital.

Procarbazine with alcohol and some foods

Some foods and drinks can cause a reaction if you are taking procarbazine. Your doctor, nurse or pharmacist can give you advice on types of food and drink to avoid.

Signs of a reaction include:

- feeling sick
- headaches
- sweating and flushing
- drowsiness
- dizziness
- a rash, which may be itchy
- sudden rise in blood pressure.

A reaction is not common. But if you notice any of these symptoms, contact the hospital for advice.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common. You may have some side effects that we have not listed here. However, in general, people manage this treatment well.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Most common side effects

These side effects are likely to be the most common side effects for this combination treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia. You will have regular blood tests during treatment to check the level of white blood cells.

An infection can be very serious when the number of white blood cells is low. Your doctor may give you antibiotics and other drugs to take during this treatment. These may help reduce your risk of getting an infection.

It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling sick

Your doctor, nurse or pharmacist will prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Tummy pain

You may get pain in your tummy (abdomen) or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help.

Steroids can irritate the stomach lining. Your doctor, nurse or pharmacist may prescribe drugs to help reduce stomach irritation.

If you have sudden or severe tummy pain, contact the hospital straight away on the 24-hour number.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Hair loss

Your hair will get thinner. Or you may lose all the hair from your head. If you want to cover up hair loss, there are different ways you can do this. Your nurse will give you information about [hair loss](#) coping with hair loss.

Remember to [protect your skin from the sun](#). Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

Skin changes

Chemotherapy can affect your skin. It might feel dry. You may develop a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Bladder irritation

Cyclophosphamide may irritate your bladder and cause discomfort when you pass urine (pee). Drink plenty of fluids every day you are taking cyclophosphamide tablets. Try to drink at least 2 litres (3½ pints).

Contact the hospital straight away if you feel any discomfort or stinging when you pass urine, or if you notice any blood in it.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Mood and behaviour changes

Steroids can affect your [mood and behaviour](#). They can cause:

- feelings of [anxiety](#) or restlessness
- mood swings (moods that go up and down)
- a low mood or [depression](#).

If you notice any changes in your mood or behaviour, tell your doctor, nurse or pharmacist. They may make some changes to your treatment if the side effects are causing you problems.

Allergic reaction

This treatment might cause an allergic reaction. But this is not common. Signs of a reaction can include:

- a skin rash or itching
- feeling breathless or wheezy
- swelling of your face, mouth or throat.

If you feel unwell or have any of these signs, contact the hospital straight away on the 24-hour number. Do not take any more of this treatment until you have checked with them.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

Effects on the heart

This treatment can affect how the heart works. You may have tests to check how well your heart is working. These may be done before, during and after treatment.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

-

breathlessness

- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Bone thinning (osteoporosis)

Steroids can cause bone thinning. Your doctor may prescribe drugs to protect your bones. They may give you advice about taking vitamin D supplements and your diet.

Doing regular exercise, such as walking, can improve your bone health. Eating a healthy diet can help too. We have more information about [bone health and looking after your bones](#).

Tell your doctor, nurse or pharmacist if you have any pain in your bones, such as in your lower back.

Eye changes

Steroids can cause eye problems. There is also an increased risk of eye infections. Tell your doctor, nurse or pharmacist if you notice any changes to your eyes such as:

- blurred vision or other vision changes
- pain
- redness.

Effects on blood pressure

This treatment can cause high blood pressure. Tell your doctor, nurse or pharmacist if you have ever had any problems with your blood pressure. You will have regular blood pressure checks during your treatment.

Other important information

Steroid emergency card

Your doctor, nurse or pharmacist may give you a steroid emergency card. You should always carry this card with you. In an emergency, it shows any doctor treating you that you are having steroid treatment. The doctor needs to know this because:

- stopping your steroid treatment or missing a dose may make you unwell
- your body may not naturally make extra steroids that are needed during a serious illness or after an injury.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Lactose

Some types of steroid tablet may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Changes to periods

If you have periods, these may become irregular or stop while you are having this treatment. They might return after treatment, but this does not always happen. Your [menopause](#) may start sooner than it would have done. Your doctor, nurse or pharmacist can give you more information.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex during this treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.
This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)